one another. The discovery of this state of matter, a kind of quark cauldron, was reported for the first time in 2002 by a team of physicists at the Brookhaven National Laboratories, Long Island, New York.

Strong theoretical evidence suggests that an episode in the very early universe, perhaps during one of the force splits, endowed the universe with a remarkable asymmetry, in which particles of matter barely outnumbered particles of antimatter: by a billion-and-one to a billion. That small difference in population would hardly get noticed by anybody amid the continuous creation, annihilation, and re-creation of quarks and antiquarks, electrons and antielectrons (better known as positrons), and neutrinos and antineutrinos. The odd man out had oodles of opportunities to find somebody to annihilate with, and so did everybody else.

But not for much longer. As the cosmos continued to expand and cool, growing larger than the size of our solar system, the temperature dropped rapidly below a trillion degrees Kelvin.

✴

A millionth of a second has passed since the beginning.

✴

This tepid universe was no longer hot enough or dense enough to cook quarks, and so they all grabbed dance partners, creating a permanent new family of heavy particles called hadrons (from the Greek *hadros*, meaning “thick”). That quark-to-hadron transition soon resulted in the emergence of protons and neutrons as well as other, less familiar heavy particles, all composed of various combinations of quark species. In Switzerland (back on Earth) the European particle physics collaboration[†] uses a large accelerator to collide beams of hadrons in an attempt to re-create these very conditions. This largest machine in the world is sensibly called the Large Hadron Collider.

The slight matter–antimatter asymmetry afflicting the quark–lepton soup now passed to the hadrons, but with extraordinary consequences.

As the universe continued to cool, the amount of energy available for the spontaneous creation of basic particles dropped. During the hadron era, ambient photons could no longer invoke $E = mc^2$ to manufacture quark–antiquark pairs. Not only that, the photons that emerged from all the remaining annihilations lost energy to the ever-expanding universe, dropping below the threshold required to create hadron–antihadron pairs.